

ETP Group 41 Project Final Report

MFB2102 Engineering Team Project II

Title	Air Regulation IoT Embedded System (ARIES)		Group Number	41
Team Name	G41			
Team Members	The team members are as follows:			
	No.	Name	ID	Dept.
	1	Levinesh Sreedharan	22012150	EEE
	2	Nurin Dyana Binti Mohd Ismail	22011877	CHE
	3	Mah Enn Chee	22011830	ME
	4	Enoch Seli Anak Maurice Medan	24006650	CHE
	5	Muhammad Fawwaz Bin Mohd Hazrizal	24006258	PE
	The leader of the team is Levinesh Sreedharan			
Coach	Dr Muslich Hartadi Sutanto			
Problem Statement & Context (Discover Phase)				
Demonstrate understanding of the real industry problem and the larger context in which the problem exists.				
Introduction	Large-scale facilities such as university hostels, office buildings, lecture halls, and commercial spaces often experience two major operational problems which are poor indoor air circulation and unnecessary energy consumption. Conventional ventilation systems commonly operate using manual switches or fixed schedules, regardless of whether a room is occupied or whether ventilation is actually required.			
	This results in stagnant indoor air, discomfort among occupants, and excessive electricity consumption caused by ventilation systems operating in empty rooms. Facility managers also have limited access to real-time information regarding room occupancy, ventilation performance, and energy usage. Moreover, students also leave their rooms without switching off their fans or lights which is a common energy sustainability issue.			
	To address these issues, the team developed ARIES: Air Regulation IoT Embedded System, an intelligent ventilation and energy-monitoring solution. ARIES automatically regulates airflow according to occupancy and environmental conditions while providing real-time energy-consumption data through a centralized dashboard.			
Background Research	Industry Insights:			
	Buildings contribute significantly to overall electricity consumption, particularly through ventilation, cooling, lighting, and other continuously operating systems. In large facilities, ventilation systems are frequently controlled using fixed timers or manual switches. These approaches are simple but inefficient because they cannot respond to real-time room conditions. Indoor air quality is also an important concern in densely occupied buildings. Poor ventilation can lead to the accumulation of carbon dioxide, humidity, odours, and volatile organic compounds. This may reduce occupant comfort, concentration, productivity, and general wellbeing. The team’s survey of 100 residents identified two important findings: 47% of respondents frequently experienced stuffy or poorly ventilated environments. 84% expressed a need for improved energy efficiency.			

	These findings indicate a clear demand for a system that improves indoor comfort while reducing unnecessary electricity consumption. Existing Solutions and Approaches: Existing ventilation solutions generally include: • Manually operated windows or ventilation flaps. • Exhaust fans controlled using basic wall switches. • Ventilation systems operating through fixed timers. • Motion-sensor systems using only PIR sensors. • Camera-based occupancy-monitoring systems. • Centralized Building Management Systems used in larger commercial buildings. Although these solutions provide some level of ventilation or automation, many are unable to accurately determine whether a room is occupied, particularly when occupants remain stationary. Advanced Building Management Systems may provide better control, but they are often expensive and difficult to install in existing facilities. User Needs and Pain Points: The research identified several important user needs and operational pain points. 1. Occupants require: • Better airflow and a more comfortable indoor environment. • A ventilation system that does not switch off while they remain stationary. • Protection from excessive humidity, outdoor pollution, or external air. • A system that does not use cameras or collect personal information. 2. Facility managers require: • Real-time monitoring of ventilation and energy consumption. • Reliable occupancy information. • Reduced electricity and operating costs. • A scalable system that can be installed across multiple rooms. • Centralized control through an accessible dashboard. Gaps and Limitations in Current Solutions: Several limitations were identified in existing ventilation and occupancy-management systems. First, manual and timer-based systems operate without considering actual occupancy or environmental conditions. This causes ventilation equipment to operate unnecessarily in empty rooms. Second, systems relying only on PIR sensors may fail to detect stationary occupants. This can cause ventilation systems to shut down even though someone is still present. Third, camera-based systems may provide more accurate occupancy information but introduce privacy concerns, higher implementation costs, and additional data-management requirements.
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	Fourth, many conventional ventilation systems do not provide real-time energy-consumption information. Facility managers are therefore unable to identify inefficient usage patterns or accurately measure energy savings. ARIES addresses these gaps by combining PIR and mmWave sensors, environmental-condition monitoring, airflow control, energy tracking, and IoT connectivity in a single modular system.
Stakeholder Identification	Stakeholder mapping was conducted to identify the key parties involved in the implementation of ARIES and understand their level of influence and interest. The stakeholders were categorized into four groups based on the Influence–Interest Matrix. 1. University Administration and Facilities Management have high influence but lower direct interest. They expect ARIES to reduce electricity costs, support sustainability targets, provide a reasonable return on investment, and integrate smoothly with existing hostel facilities. 2. Hostel Residents or Students are the selected Extreme Users because they have both high interest and high influence. They are the main users who directly experience poor ventilation, overheated rooms, and the inconvenience due to not having a proper ventilation system in their rooms. They expect ARIES to provide improved thermal comfort, reliable occupancy detection, automatic operation, and lower energy waste without affecting their daily routine. Their feedback is especially important because it determines whether the system is comfortable, practical, and acceptable for real hostel use. 2. Maintenance Staff and the Facilities Team have high interest but lower influence. They expect the system to be easy to install, inspect, repair, and maintain. They are also concerned about component reliability, system downtime, and the availability of clear fault indicators. 3. Visitors and Parents of Students have low influence and low interest. They expect the hostel environment to be safe, comfortable, and well maintained, but they are not directly involved in the operation or management of ARIES. Attachment: Stakeholder Mapping

	 Extreme User (High Interest – High Influence) The selected Extreme User is Hostel Residents or Students, as they interact with the system daily and directly experience its impact on indoor comfort and energy efficiency. Students often face issues such as poor ventilation, overheated rooms, and inconvenience due to not having a proper ventilation system in their rooms. They expect an intelligent solution that automatically regulates airflow, improves living comfort, and minimizes unnecessary energy consumption. Their feedback is essential in evaluating the effectiveness and usability of ARIES.
Personas	The selected persona represents the primary target user of ARIES and was developed based on user observations and survey findings. Ramli bin Suden is a 21-year-old second-year Petroleum Engineering student living in Village 5, Universiti Teknologi PETRONAS. As a busy student, he values convenience and prefers systems that operate automatically with minimal manual intervention. He believes that unnecessary energy consumption should be avoided and prioritizes practical, reliable solutions over luxury. Ramli frequently forgets to switch off electrical appliances due to his demanding academic schedule and often experiences thermal discomfort caused by poor room ventilation. He believes that an integrated smart ventilation system capable of automatically regulating airflow and reducing energy wastage would significantly improve both comfort and quality of life in the hostel. This persona guided the development of ARIES by ensuring that the solution focuses on automation, energy efficiency, user convenience, and improved indoor air quality. Attachment: Personas

	Name Ramli bin Suden	Age 21	
	Gender Male	Income 0	Education Petroleum Engineering
	Social Roles UTP Student 2<sup>nd</sup> Year 2<sup>nd</sup> Sem living in Village 5		
	Values, belief and principles • Values convenience, simplicity and prefer system that work automatically • Beliefs energy waste is undesirable • His holds to practicality over luxury and reliability	Perspectives on issue • Often forgets to turn off appliances due to busy schedule and cognitive load • Thermal discomfort due to overheated room • Lack of automation system • Believes an intergrated system that required minimal manual interaction would really save the energy and better life experience	

Problem Definition (Define Phase)

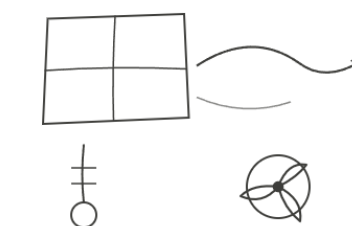
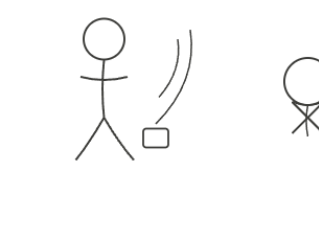
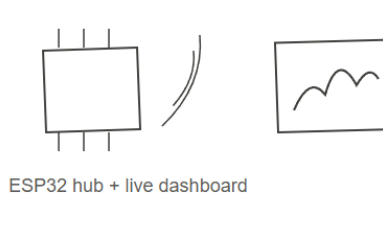
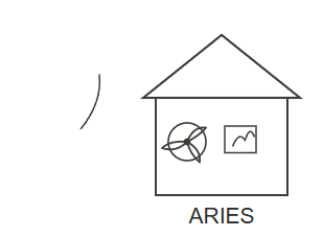
Narrow down and clearly articulate the specific challenge the team are addressing.

Opportunity Statement	A focused, well-defined How-might-we statement that the team agreed on.
Key Criteria for Success	Clear, measurable criteria that will determine the success of their solution. These should include both technical and user-centered considerations. What does "Success" look like?

Ideation & Concept Development (Develop Phase)

Explore multiple potential solutions, demonstrating creative problem-solving and feasibility analysis.

Collaborative Sketching	With our opportunity statement in mind, we used the collaborative sketching method to brainstorm solutions to poor indoor air quality and energy inefficiency. Our top 3 ideas that the team voted for were: 1. Smart Ventilation System – a 12V DC ventilation fan that automatically turns on/off based on indoor vs. outdoor temperature comparison, reducing excessive reliance on air-conditioning. 2. Occupancy-Based Energy System – using dual-sensor fusion (PIR motion + mmWave presence detection) to cut power to unnecessary loads only when a room is genuinely empty, while still recognizing stationary/resting occupants. 3. Central IoT Controller with Live Dashboard – an ESP32-based hub that consolidates sensor data and displays real-time energy usage, tariff cost, and occupancy status to the user. Through the C-Sketch process, these three ideas were merged into a single integrated concept rather than treated as separate products, the ventilation logic was linked to temperature-difference triggering, the occupancy detection was refined with mmWave to solve the "false-off during rest" problem common in motion-only systems, and everything was tied together through the central microcontroller and dashboard. This resulted in the final concept: ARIES (Air Regulation IoT Embedded System).
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	1. Smart ventilation  temp-triggered vent + fan 2. Occupancy energy  PIR + mmWave, load auto-off 3. IoT controller  ESP32 hub + live dashboard 4. ARIES (merged)  ARIES vent + occupancy + IoT — one system
Real-Win-Worth	• Real: Our survey of 100 respondents confirmed the problem is genuine, 47.5% reported frequently stuffy rooms, and 84% rated energy-efficient regulation as important, yet most facilities still rely on manual, always-on ventilation. • Win: ARIES's advantage lies in dual-sensor fusion (PIR + mmWave), which reliably distinguishes an empty room from a stationary occupant, a gap in conventional motion-only systems, without using privacy-invasive cameras. • Worth: The system uses low-cost, readily available components (ESP32, PIR/mmWave sensors, relay modules), keeping unit cost low while our testing showed an 85% reduction in ventilation operating costs and 100% ROI within the first year, making it both technically and economically justified.
Solution & Development (Develop Phase) <i>Outline the project's solution and development.</i>	
Detailed Solution Overview	ARIES is an automated, occupancy-based ventilation and energy-tracking system designed for enclosed indoor spaces such as hostel rooms. The system's operating principle is simple: only run ventilation and power loads when a room is occupied and conditions call for it, cutting the energy waste of conventional always-on or timer-based systems. <i>Functional Requirements:</i> 1. Detect room occupancy reliably, including stationary/resting occupants. 2. Automatically control airflow based on occupancy and environmental conditions using a fan mounted on the window. 3. Monitor real-time power consumption (current, voltage, energy, estimated cost). 4. Present live system status through a remote dashboard.

	Component	Function
	ESP32	Main controller + Built in WiFi module — reads sensors, evaluates control logic, drives relays and receives sensor data serially from the microcontroller and uploads it to the live web dashboard.
	PIR motion sensor	Detects active occupancy (movement)
	mmWave sensor	Detects both motion and static presence of human
	DHT11 sensor	Measures indoor temperature and humidity
	INA219 current/voltage sensor (I2C)	Measures real-time power draw of connected loads
	Relay Module	Controls the DC fan (ventilation) and LED light load
	Power Adapter	Powers the whole system with a 12V DC stable power source.
	Buck Converter	Converts the 12V DC to stable 5V DC power source suitable for ESP32 microcontroller.
	Jumper wires & Breadboard	Connects the components to the microcontroller
	OLED Screen	Display live data to users/residents.
	Live dashboard (web/mobile)	Displays temperature, humidity, occupancy status, power (W), energy (Wh/kWh), estimated cost, and system status

System Logic Flow:

- Occupancy Detection** — PIR and mmWave/presence sensors continuously check whether the room is occupied.
- If occupied:** the system monitors energy usage (current, voltage, power), analyzes consumption to identify high-usage patterns, and controls devices accordingly (relay ON, fan/light activated as needed).
- If unoccupied:** the system enters energy-saving mode — unnecessary loads are switched off (relay OFF) and the system enters standby monitoring.
- In both cases, the dashboard is updated in real time with power/energy data and estimated savings, and the cycle repeats continuously.

Power and Signal Design:

The prototype runs on a DC-based low-voltage architecture for lab safety and ease of testing: a 12V DC adapter supplies the fan and light loads directly, while a buck converter steps this down to 5V for the ESP32 and sensors. Sensor signals (low current) are wired separately from the relay-switched load lines (high current) to reduce interference and wiring risk. This design was selected for safety at prototype voltage, ease of build and testing, and confirmed suitability for lab demonstration, while remaining scalable toward a real-world deployment.

Project Milestones													
Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	
1													
2													
3													
4													
5													

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Roles and Responsibilities	Clear division of tasks among team members based on their disciplines (mechanical, electrical, civil, chemical, etc.). <table><tr><th>No.</th><th>Name</th><th>Role</th></tr><tr><td>1</td><td>Levinesh Sreedharan</td><td>Responsible for firmware development, hardware integration, energy-saving logic on the ESP32 and debugging</td></tr><tr><td>2</td><td>Nurin Dyana Binti Mohd Ismail</td><td>Testing and validation of the prototype trials, sensor sensitivity tuning, data collection for results.</td></tr><tr><td>3</td><td>Mah Enn Chee</td><td>Mechanical design of the structure using AutoCAD & building the web dashboard for live data display & tracking</td></tr><tr><td>4</td><td>Enoch Seli Anak Maurice Medan</td><td>Testing and validation of the prototype trials, sensor sensitivity tuning, data collection for results</td></tr><tr><td>5</td><td>Muhammad Fawwaz Bin Mohd Hazrizal</td><td>Building of the physical structure of the prototype & deciding the material to make the final structure</td></tr></table> List down 3 principles (ground rules) that the team subscribed to for this project. 1. Every member takes ownership of their assigned subsystem and communicates blockers early rather than letting them surface at integration. 2. Decisions on design changes are discussed and agreed as a team before implementation, especially where one subsystem affects another. 3. All members contribute to testing and documentation, not just their own component, no single person carries the reporting workload alone.	No.	Name	Role	1	Levinesh Sreedharan	Responsible for firmware development, hardware integration, energy-saving logic on the ESP32 and debugging	2	Nurin Dyana Binti Mohd Ismail	Testing and validation of the prototype trials, sensor sensitivity tuning, data collection for results.	3	Mah Enn Chee	Mechanical design of the structure using AutoCAD & building the web dashboard for live data display & tracking	4	Enoch Seli Anak Maurice Medan	Testing and validation of the prototype trials, sensor sensitivity tuning, data collection for results	5	Muhammad Fawwaz Bin Mohd Hazrizal	Building of the physical structure of the prototype & deciding the material to make the final structure																																															
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Project Implementation (Deliver Phase) <i>Discuss the project development, testing, iteration and final product.</i>																																																																		
Prototype Development	The prototype development began with building the room structure using acrylic board. The team used the university's laser cutter to cut the acrylic board into pieces of the prototype's wall. The electronic components, including motion sensors, a																																																																	

	microcontroller ESP32, LED indicators, and the fan, were assembled in the electronics lab. Initial challenges included difficulty in calibrating the sensors to detect motion at an optimal distance and integrating all components into a compact design. These challenges were addressed through a circuit simulation software named “Circuit Designer”. We conducted multiple iterations and trials in the simulation software , where the sensor placement and wiring configuration were refined as well as the coding for the sensors.
Testing and Iterations	Once the initial prototype was ready, we tested it within the campus, particularly in the Village 5 common room. We wanted to see the motion sensors in action as well as the temperature sensor to detect about 2 degrees of temperature differences. From the result of our testing, it appears that our sensors were too sensitive towards the motion of human within 1 meter of distance as well as a slight error where the LED indicators responded to the vacancy of the room instead of occupancy. We modified the codes of command to do the opposite and tried to make the temperature sensor to work faster by using heat of hair curler as a means of demonstration. 
Final Design	The final design of the miniature ARIES system is combining PIR with high-precision mmWave radar with a live dashboard display for the data such as the watt usage and the live current and energy consumption. The fan acting as the representation of the vent is connected to the temperature sensor only. As the sensitivity of the mmWave sensor is too high, we have to demonstrate it by ensuring that nobody is within 1 meter distance from the sensors.
Result and Impact <i>Deliberate the result gained from the solution implementation.</i>	
Results	The ARIES system was implemented across dormitory test units to evaluate its ventilation performance, energy savings, and operational safety. The data collected showed an 85% reduction in annual ventilation operating costs, dropping expenses from RM 150–200 down to RM 20–30 per unit. Additionally, the system achieved a 100% Return on Investment within the first year by lowering energy draw from 60–100 Watts down to 5–15 Watts during occupied periods. These results demonstrate the effectiveness of ARIES in reducing building energy waste while maintaining proper indoor air quality. The effectiveness of the system hinges on the precision of the dual-sensor fusion, which combines PIR and mmWave radar for reliable occupancy detection. The

	stability of the Wi-Fi connection to the central dashboard is crucial for real-time energy monitoring and automated vent control. Built-in safety features, such as low-voltage operation and emergency alarm integration, played a vital role in ensuring user safety and compliance. The results demonstrate the potential of the system to be scaled to larger spaces, including lecture halls and office building.
User Feedback	From an industrial investor viewpoint, they expressed satisfaction with the ease of use and effectiveness of the ventilation system. They noted that the automation and the dashboard really help ease the monitoring for a large scale implementation. They also suggest including a cooling mechanism to make the product more marketable in public eye.
Environmental, social, and governance (ESG) Impact	• Environmental: Traditional vents waste electricity running on non-stop timers, even in empty rooms. ARIES fixes this by turning on only when people are actually present, slashing unnecessary power draw and reducing greenhouse gas emissions. • Social: It keeps people healthier by cycling in fresh air to prevent stale air buildup, headaches, and sickness—all without invading anyone's privacy since it uses radar sensors instead of cameras. • Governance: It replaces guesswork with real data. Facility managers get a central dashboard showing exact energy use, built-in safety features, and auto-open mechanisms during emergencies, making it easy to prove compliance with environmental regulations and safety standards.
Challenges and Lessons Learned <i>Outline the ups and downs while executing this project.</i>	
Technical Challenges	One of the primary technical challenges was calibrating the mmWave sensor to accurately detect human presence. During initial testing, the sensor occasionally produced false detections or missed occupants, particularly when they remained stationary for extended periods. To address this issue, the team conducted multiple rounds of testing and adjusted the sensor's detection range, sensitivity, and placement to improve accuracy and reliability. Another challenge involved integrating data from multiple sensors, including the mmWave sensor, PIR motion sensor & temperature sensors into a single control system. Ensuring smooth communication between the sensors and the microcontroller requires careful programming, debugging, and system optimization. Through iterative testing and software refinement, the team successfully achieved stable sensor integration and reliable automated system operation.
Teamwork and Collaboration	Throughout the project, the team encountered challenges in coordinating tasks, as each member was responsible for different components, including hardware integration, sensor programming, prototype construction, and system testing. This occasionally led to delays in combining individual work into a fully functional system. To improve collaboration, the team held regular progress meetings and maintained continuous communication through a WhatsApp group. Tasks and deadlines were clearly assigned, allowing members to share updates, discuss technical issues, and provide support when needed. This collaborative approach enabled the team to identify and resolve problems more efficiently, resulting in smoother system integration and timely completion of project milestones.

Budget and Resource Allocation

Tabled the expenses for this project.

Expenses breakdown

Here is the breakdown for the budget utilisation

Item	Price (RM)
Microcontroller (ESP32)	35
Sensor Suite (PIR Motion x1, DHT11 Temp/Humidity x2, mmWave radar sensor)	60
Relay Control Modules (Power Cutoff & Fan Switching)	15
Air Ventilation Components (DC Cooling Fan)	20
Acrylic Material & Enclosure Casing	60
Power Adapters and Wiring	40
Prototype Testing & Miscellaneous (Fasteners, Adhesives, Connectors)	50
Buck Converter + DC Jack Adapter	20
Breadboard + Jumper Wires	10
INA219 Current Sensor	10
OLED Display	20
Cloud Dashboard Integration	60

Commercial Market

Given:

Cost = RM400

Profit Margin = 50%

Profit Calculation:

Profit = (Profit Margin / 100%) * Cost

Profit = (50% / 100%) * RM400

Profit = 0.5 * RM400

Profit = RM200

Selling Price:

Selling Price = Cost + Profit

Selling Price = RM400 + RM200

Selling Price = RM600

	If the smart room automation and air quality management system were to be commercialized, the estimated selling price would be around RM600 per unit. This estimate includes a margin for profit and covers production costs such as sensors, microcontrollers, relays, labor, and assembly. In a larger-scale production run, bulk sourcing of IoT components could significantly reduce unit costs, allowing the selling price to be even more competitive in the smart home and eco-friendly automation market.
Sustainability and Future Recommendations <i>Ensuring Long-Term Sustainability and Development.</i>	
Scalability	➤ Multi-Zone Expansion via ESP-NOW Wireless Mesh The prototype currently manages a single room, but its design(sensors, relay control, and automation logic) can be replicated across multiple rooms. Using ESP-NOW, a peer-to-peer wireless protocol built into the ESP32, additional room nodes could report occupancy, temperature, and consumption to a central aggregator without any router infrastructure, turning the system into a building-scale energy management platform. This would help us to scale it into a whole hostel block in a Residential Village in UTP, which could then eventually be scaled into all village blocks in UTP. ➤ Per-Branch Energy Monitoring Consumption is currently measured by a single INA219 current sensor on the main supply and attributed to appliances from known relay states. Adding one sensor per switched branch, as the INA219 supports multiple I2C addresses, would enable true per-appliance measurement, billing, and health monitoring which is increasingly valuable as the number of controlled loads grows.
Improvements and Future Work	• Solar Generation and Net Energy Metering (NEM) Integration Since the billing engine already models the national tariff, extending it to Malaysia's NEM scheme, where exported solar energy earns bill credits is a natural step. A solar-maintained battery would let the dashboard show generation offset alongside consumption cost, strengthening both the awareness function and the existing outage-resilience feature, in line with SDG 7. • AI-Driven Predictive Occupancy Pre-Conditioning With occupancy timestamps logged over time, a machine learning model could learn the room's recurring occupancy rhythms which includes daily and weekly patterns, including their variability and pre-condition the room shortly before predicted return times, rather than reacting only after presence is detected. Lightweight TinyML frameworks allow simple pattern models to run directly on ESP32-class hardware. • True Per-Appliance Load Disaggregation (NILM) Non-Intrusive Load Monitoring which helps to identify operating appliances from the aggregate electrical signal alone would extend consumption awareness to plug loads outside the system's direct control. As NILM requires high-frequency sampling and signature-classification algorithms, it is positioned as long-term research-grade work rather than an incremental step. This step requires some deep research before integrating this feature into our current system.

	Maintenance-Safety Digital Interlock Fatal accidents occur when a circuit de-energized for maintenance is unknowingly re-energized by another person. A future version could implement a software-defined interlock, where a physically-engaged maintenance switch causes the controller to reject all automated and remote re-activation commands until released at the panel, with each event logged to non-volatile storage. This does not replace physical lockout-tagout for hazardous work, but closes the specific gap of remote or automated re-energization. Hardware Robustness: From Breadboard to Permanent Construction During development of the prototype, we faced recurring breadboard contact issues such as intermittent I2C failures, sensor dropouts after nearby wiring changes, and boot-time faults requiring firmware mitigation. A deployed system with soldered or PCB construction, proper connectors, and decoupling of the switching supply from signal paths would be a more standardized way to represent our prototype for industry ready adaptation.
Conclusion <i>Summarize the key points and outline the immediate next steps post-proposal submission.</i>	
Closing of the project	We began this project with a simple observation in hostels where lights and fans run in empty rooms all the time, and nobody sees what it costs until the bill arrives. Our goal was to build a room that could manage its own energy use and, just as importantly, show its occupant what that use actually costs in ringgit. Looking back at what we set out to do, we believe the project met its goals. The room we built switches its own lighting and ventilation based on whether anyone is actually present, using a PIR motion sensor together with an mmWave radar sensor so that even someone sitting still is not mistaken for an empty room. The fan goes further than simple occupancy, in which, it only runs when the outdoor air is genuinely cooler than indoors by a meaningful margin, so it never wastes energy pushing warm air around. At the same time, the resident always stays in control through a manual override, because automation should assist people, not overrule them. What we are most proud of is the metering side. Instead of assuming appliance wattages like many similar projects do, our system measures real current and runs it through a billing engine built on TNB's actual RP4 tariff which includes tiered rates, surcharges, service tax and all with a meter register that never resets, just like a real one. Along the way, we added things we had not originally planned such as an automation that becomes stricter as consumption crosses tariff price boundaries, an in-room display showing live cost, a phone dashboard and a counter that shows how much energy the automation has saved. We also want to be honest about what needs refinement. The radar still needs proper calibration to adapt to a UTP hostel room as currently it was adapted for a prototype, the demonstration runs on accelerated time and scaled power, and building on breadboards taught us some hard lessons about loose connections that a soldered version would avoid. None of this changes what the prototype proves.

	As for the hostel energy wastage problem itself, our contribution comes down to two things. The system removes the need for anyone to remember to switch things off as the room simply knows when it is empty. And it closes the awareness gap that research on hostel energy use keeps pointing to, as for the first time, a resident can watch, in real time, what their room is consuming and what has been saved. Wasted energy in student housing is mostly a human problem, and we tried to solve it by pairing automation with awareness. On the evidence of this prototype, that pairing works and it is ready to grow beyond one room.
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Appendices

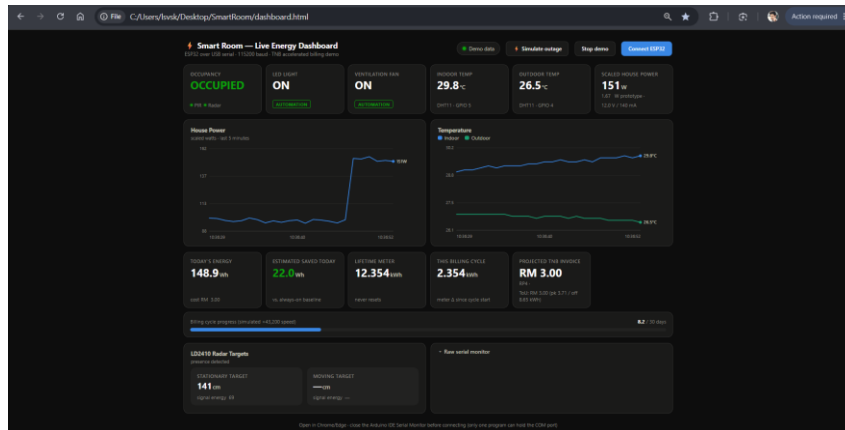
References	➤ Ng, T. F., Ahmad Shabudin, A. F., Hassan, S., [fifth author — see note], & Ibrahim, K. (2017). Energy consumption in student hostels of Universiti Sains Malaysia: Energy audit and energy efficiency awareness. In W. Leal Filho, M. Mifsud, C. Shiel, & R. Pretorius (Eds.), <i>Handbook of theory and practice of sustainable development in higher education</i> (Vol. 3, pp. [page range]). Springer. https://doi.org/10.1007/978-3-319-47895-1_12 ➤ How2Electronics. (2025). <i>LD2410 sensor with ESP32 – Human presence detection</i>. https://how2electronics.com/ld2410-sensor-with-esp32-human-presence-detection/ ➤ Santos, R., & Santos, S. <i>ESP32 pinout reference: Which GPIO pins should you use?</i> Random Nerd Tutorials. https://randomnerdtutorials.com/esp32-pinout-reference-gpios/ ➤ Santos, R., & Santos, S. <i>ESP32 relay module – Control AC appliances (web server)</i>. Random Nerd Tutorials. https://randomnerdtutorials.com/esp32-relay-module-ac-web-server/ ➤ Tan, P. (2025, June 20). <i>Big changes to domestic TNB bill structure – no more tiered rates, ToU option for domestic, from July 1, 2025</i>. Paul Tan's Automotive News. https://paultan.org/2025/06/20/big-changes-to-domestic-tnb-bill-structure-no-more-tiered-rates-base-tariff-up-by-13-64-from-july-2025/ ➤ Cirkuit Designer. <i>How to use INA219: Pinouts, specs, and examples</i>. https://docs.cirkitdesigner.com/component/b753512c-f916-431c-9e71-325ec6038562 ➤ Sri, K. <i>IOT-based-Smart-buildings</i>. GitHub. https://github.com/KushagraSri87/IOT-based-Smart-buildings ➤ Tenaga Nasional Berhad. (2025, July). <i>Engagement with RTMT customer on tariff restructuring RP4</i> [Presentation slides]. Malaysia Retailers Association. https://mra.com.my/wp-content/uploads/2025/07/New-RP4-Electricity-Tariff-Structure-Now-in-Effect-%E2%80%93-What-Retailers-Should-Know.pdf ➤ Zhang, J., Lutes, R. G., Liu, G., & Brambley, M. R. (2013, January). <i>Energy savings for occupancy-based control (OBC) of variable-air-volume (VAV) systems</i> (PNNL-22072). Pacific Northwest National Laboratory.
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https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-22072.pdf

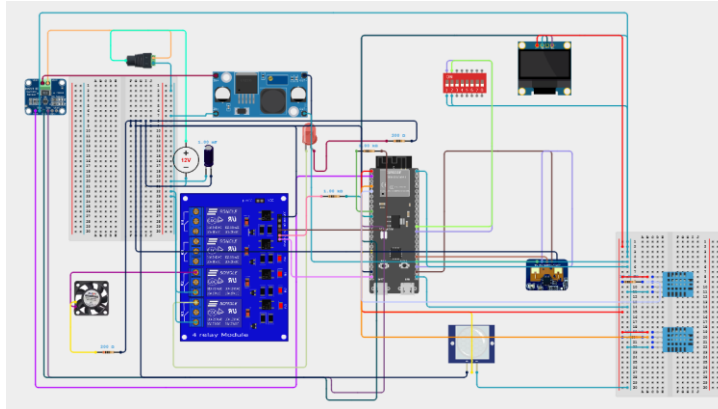
- Nilsson, A., Wester, M., Lazarevic, D., & Brandt, N. (2018). Smart homes, home energy management systems and real-time feedback: Lessons for influencing household energy consumption from a Swedish field study. *Energy and Buildings*.
<https://www.sciencedirect.com/science/article/abs/pii/S0378778818311691>

Additional Visuals/ Diagrams

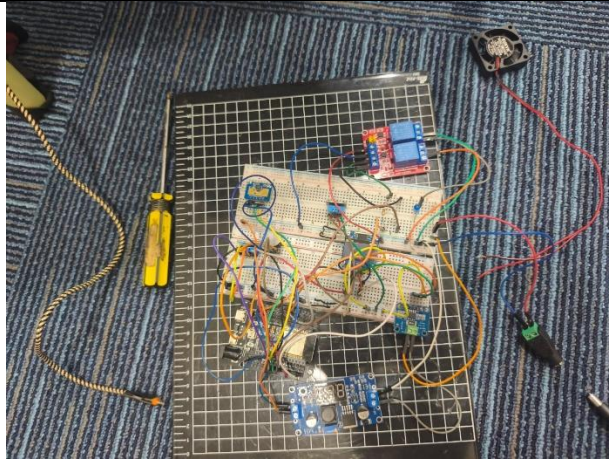
- Web dashboard created for the prototype



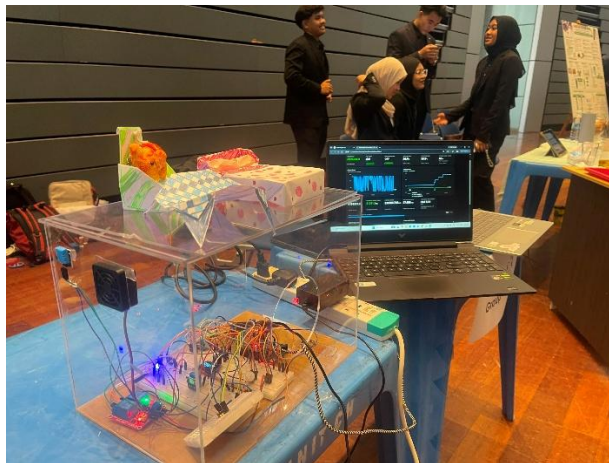
- Circuit Simulation using Cirkuit Designer software



- Circuit connection done by referring the circuit schematics



➤ Image of final prototype presentation during ETPx



➤ Prototype testing by judges & industry expert

